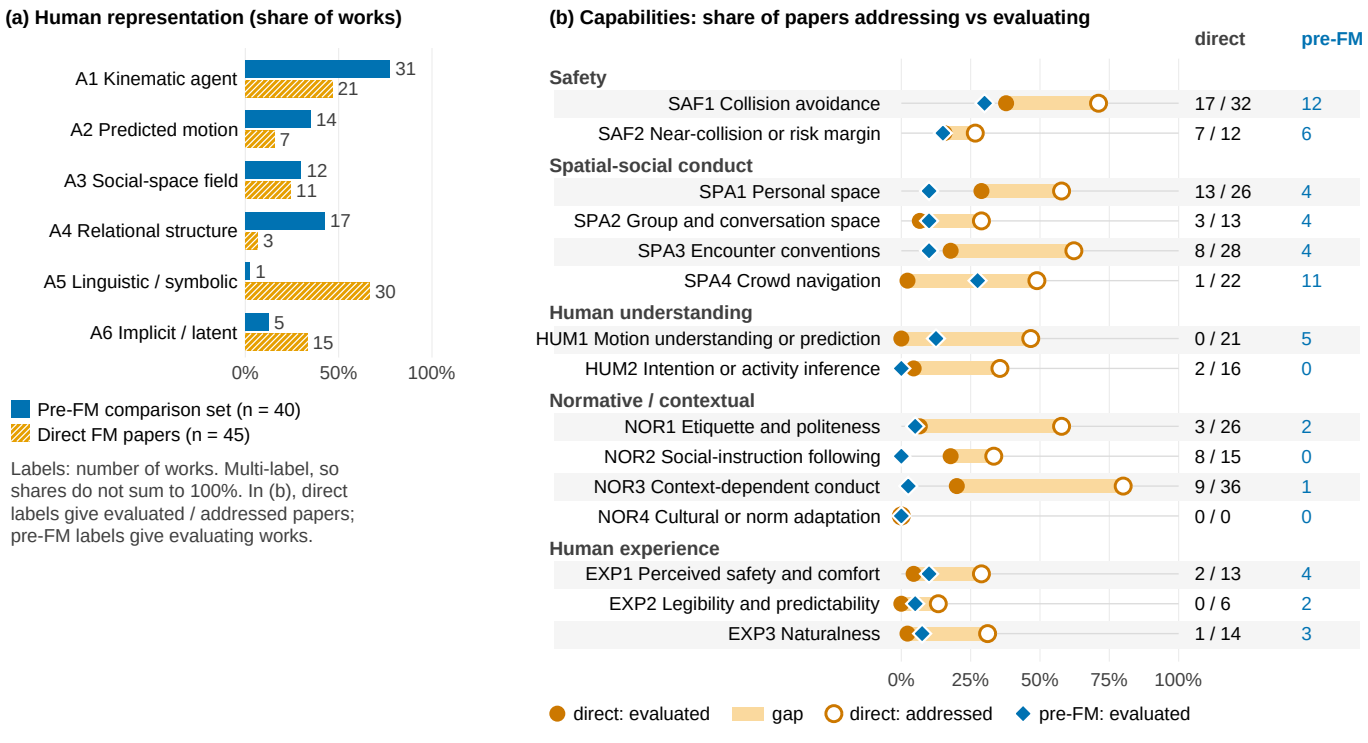
(a) Human representation (share of works)
A1 Kinematic agent
31
21
A2 Predicted motion
14
7
A3 Social-space field
12
11
A4 Relational structure
17
3
A5 Linguistic / symbolic
1
30
A6 Implicit / latent
5
15
0%
50%
100%
Pre-FM comparison set (n = 40)
Direct FM papers (n = 45)
Labels: number of works. Multi-label, so shares do not sum to 100%. In (b), direct labels give evaluated / addressed papers; pre-FM labels give evaluating works.
(b) Capabilities: share of papers addressing vs evaluating
direct
pre-FM
Safety
SAF1 Collision avoidance
17 / 32
12
SAF2 Near-collision or risk margin
7 / 12
6
Spatial-social conduct
SPA1 Personal space
13 / 26
4
SPA2 Group and conversation space
3 / 13
4
SPA3 Encounter conventions
8 / 28
4
SPA4 Crowd navigation
1 / 22
11
Human understanding
HUM1 Motion understanding or prediction
0 / 21
5
HUM2 Intention or activity inference
2 / 16
0
Normative / contextual
NOR1 Etiquette and politeness
3 / 26
2
NOR2 Social-instruction following
8 / 15
0
NOR3 Context-dependent conduct
9 / 36
1
NOR4 Cultural or norm adaptation
0 / 0
0
Human experience
EXP1 Perceived safety and comfort
2 / 13
4
EXP2 Legibility and predictability
0 / 6
2
EXP3 Naturalness
1 / 14
3
0%
25%
50%
75%
100%
direct: evaluated
gap
direct: addressed
pre-FM: evaluated